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EV Charging Support — Knowledge Base Index

Version: 1.0
Maintained by: Support Operations
Last Updated: May 2026

Purpose

This knowledge base is the authoritative reference for EV charging support agents. Each article maps to a specific fault category and provides diagnostic logic, agent scripts, escalation criteria, and background context. Articles are written to be usable during a live call.

Article Directory

Article	Topic	Call Volume Share
KB-001	Session Won't Start / Ended Early	~27% of contacts
KB-002	Payment & Billing Issues	~23% of contacts
KB-003	Hardware Fault vs. Vehicle-Side Issue Diagnosis	~18% of contacts
KB-004	Ghost Availability (App Shows Green, Charger Won't Start)	~8% of contacts
KB-005	CCS Charging — Full Driver Guide & Troubleshooting	~12% of contacts
KB-006	Slow or Throttled Charging	~6% of contacts
KB-007	App, Account & RFID Card Issues	~9% of contacts
KB-008	Stuck / Locked Cable	~4% of contacts
KB-009		Always active

Article	Topic	Call Volume Share
	Safety	
	Escalation	
	Procedures	

Quick Triage Guide

Driver says...	Go to
“Nothing’s happening,” “won’t start,” “shows an error”	KB-001
“It stopped on its own,” “session ended early”	KB-001
“Card declined,” “charged me but didn’t charge my car,” “wrong bill”	KB-002
“Screen is black,” “charger looks broken,” “hardware issue”	KB-003
“App says it’s available but it won’t turn on”	KB-004
“How do I use this CCS plug?”, “connector won’t fit,” “CCS issue”	KB-005
“Charging really slowly,” “barely getting any charge”	KB-006
“Can’t log in,” “RFID card won’t work,” “app problem”	KB-007
“Cable is stuck,” “can’t unplug”	KB-008
“It’s sparking,” “burning smell,” “I got a shock”	KB-009 — IMMEDIATE

Remote Action Reference

Remote Action	Authorized for	Time to effect
Soft reset	Session start failure, session drop, slow charging	~30 seconds
Hard reset	Soft reset failed, station unresponsive	~90 seconds
Connector unlock	Stuck cable	~10 seconds
Station status check	All issues — always run first	Immediate
Session credit / billing flag	Payment taken for failed or incomplete session	Logged for billing team
Nearest charger lookup	Station down, driver needs to relocate	Immediate

Remote Action	Authorized for	Time to effect
Technician dispatch	Hardware fault, stuck cable, safety concern	Per region SLA

Escalation Thresholds

Escalate to Tier 2 immediately if: - Remote reset fails twice with station showing as online - Cable is physically stuck and remote unlock fails - Driver suspects vehicle electrical damage - Billing dispute exceeds \$50 or involves fraud - Driver is stranded with no reachable alternative - Any safety concern: sparking, burning smell, smoke, electric shock

For safety emergencies, always direct the driver to disconnect and step back before any other step. If fire or shock is reported, instruct the driver to call emergency services.

KB-001 — Session Won't Start / Session Ended Early

Category: Session Issues

Call Volume: ~27% of all support contacts

Resolution Rate on Call: ~95%

Primary Remote Action: Soft reset, Hard reset

Why This Matters

Session start and premature-end failures are the single highest-volume support category. The good news: approximately 27% of these resolve with a remote reset alone, making them highly automatable. Understanding the failure chain — from driver tap to active charging — lets agents triage quickly and avoid unnecessary technician dispatches.

Background: How a Charging Session Actually Starts

Understanding the handshake helps agents reason about where in the chain a failure occurred.

1. **Driver presents payment or credential** (app, RFID card, tap-to-pay, plug-and-charge)

2. **Network authorizes the session** — the charger contacts the OCPP backend to validate payment and unlock the connector
3. **Connector lock engages** — the physical latch locks the cable into the vehicle inlet
4. **ISO 15118 / OCPP handshake** — the charger and vehicle negotiate power levels, charge limits, and session parameters
5. **Power flow begins** — DC or AC current flows; the session is live

A failure at any step produces a different symptom. If the agent knows where it broke, they skip unnecessary steps.

Section A — Session Won't Start

Common Root Causes

Cause	Frequency	Identifiable by
Loose or unseated cable connection	~15%	Resolves on re-plug
OCPP authorization timeout or backend error	~20%	Remote reset resolves
Payment/credential authorization failure	~18%	Error code on screen; see KB-002
Vehicle-side communication fault	~12%	Multiple chargers fail for same vehicle
Charger hardware fault (relay, cable, port)	~10%	Persists after reset; see KB-003
Ghost-online / stale telemetry	~8%	App shows available, charger unresponsive; see KB-004
Software/firmware crash	~7%	Hard reset resolves
RFID or app authentication issue	~10%	See KB-007

Diagnostic Script — Session Won't Start

Step 1 — Confirm setup and get context

“Can I get the station ID — it’s usually a 4 to 6 digit number on a sticker on the charger?”

“And what exactly do you see on the screen — any error codes or messages?”

“When you plugged in, did you hear a click, or did the connector feel loose?”

Step 2 — Re-seat the cable

“First thing to try: unplug the cable completely and plug it back in firmly until you hear or feel it click. Sometimes a slightly loose connection is all it takes.”

Wait 30 seconds. If the session starts — confirm it’s running, log resolution, close call.

Step 3 — Run remote station status check

Before resetting, check the station’s real-time status in the backend:

- **Station offline / faulted:** Do not reset yet. Go to KB-003 Hardware Fault.
- **Station online / available:** Proceed to soft reset.
- **Station shows “in session”:** A ghost session may be open — clear it in the backend before resetting.

Step 4 — Soft reset

“I’m sending a remote restart to the station now — it takes about 30 seconds. Please stay plugged in and tell me when the screen changes.”

Trigger soft reset on Station ID. Wait 45 seconds. If session starts — confirm it’s running, close call.

Step 5 — Hard reset (if soft reset fails)

“The soft restart didn’t do it. I’m going to try a full power cycle on the station — this takes about 90 seconds.”

Trigger hard reset. Wait 2 minutes. If session starts — confirm, log, close.

Step 6 — If both resets fail

“The station isn’t responding to resets. Let me check something — have you tried charging at any other station recently? Did that work okay?”

- **If other stations work fine:** Fault is likely on this specific charger. Go to KB-003.
- **If other stations also fail:** Strongly suggests a vehicle-side issue. > “It’s possible the issue is actually with your car’s charging port or communication system. I’d recommend contacting your vehicle

manufacturer or dealer — I know that’s not what you wanted to hear, and I’m sorry.”

Step 7 — If unresolved after 3 attempts

“I’m flagging this station for a technician inspection. I’m really sorry this didn’t get sorted today. Can I help you find the nearest available working charger while we wait?”

Offer nearest-charger lookup. Apply session credit if applicable.

Error Codes — Session Won’t Start

Display Code	Meaning	Agent Action
E001 / E01	Network communication error	Soft reset; if persistent, escalate
E002 / AUTH FAIL	Authorization/ payment declined	Go to KB-002
E003 / CONNECTOR	Connector not fully seated	Ask driver to re-seat cable
E004 / VEHICLE	Vehicle not communicating	Check vehicle-side; may be car fault
E005 / GROUND	Ground fault detected	Do not reset — safety flag, escalate
OFFLINE	Charger not on network	Hard reset; if persistent, dispatch tech
RESERVED	Station reserved by another user	Check backend for open reservation
OUT OF ORDER	Hardware fault flagged	Go to KB-003

Note: Error codes vary by manufacturer (ABB, BTC Power, ChargePoint, Tritium, Wallbox, etc.). If a code isn’t listed, log it verbatim for the technician.

Section B — Session Ended Early

Common Root Causes

Cause	Frequency	Identifiable by
Vehicle BMS charge limit reached	~25%	Car near full; driver set a cap
	~20%	Mid-session, no screen error

Cause	Frequency	Identifiable by
Grid fluctuation / thermal trip		
Payment authorization expiry	~18%	Session ends at a billing interval
Handshake dropout (ISO 15118 failure)	~15%	Screen shows “communication error”
Network/connectivity loss at charger	~10%	Screen goes dark or shows “offline”
Hardware thermal protection trip	~8%	Hot weather, extended session
Vehicle-side fault	~4%	Vehicle dashboard warning light

Diagnostic Script — Session Ended Early

Step 1 — Get timeline and symptoms

“How far into the session was it when it stopped — roughly how many minutes or what percentage?”

“Did anything show on the screen when it stopped, or did it just go quiet?”

“Did you see any warning lights on your car’s dashboard?”

Decision tree based on timing:

Under 2 minutes: Likely a payment authorization failure or handshake failure. > “An early stop like that is usually a payment authorization issue. Let me check the session on our end.” - If payment failed: Go to KB-002. - If handshake error: Soft reset and retry.

2–20 minutes in, no screen message: Likely a grid fluctuation or network dropout. > “A drop in the middle like that can be a brief power grid fluctuation or a temporary connectivity hiccup at the station. I’m going to send a reset and you can try starting a new session — if it drops again at the same point, we’ll get a tech out.”

Mid-session with a vehicle dashboard warning: Vehicle-side issue. > “That dashboard warning is your car telling you something — it may have stopped the session as a protective measure. I’d suggest checking your vehicle manual or contacting your dealer. The charger may be fine on our end.”

Session showed “complete” but battery isn’t full: Charge limit set in vehicle. > “It’s worth checking in your vehicle’s app or settings — a lot of drivers set a charge cap at 80 or 90 percent to protect their battery. If there’s a cap set, that would explain this.”

Step 2 — Retry session

“Let me send a reset to the station and you can try starting a new session. Please let me know if it drops again.”

Step 3 — If session drops a second time in the same phase

“Dropping twice in the same spot suggests a hardware or firmware issue at this station. I’m going to flag it for a technician. I’m really sorry about the inconvenience.”

Step 4 — Billing review for incomplete session

“If you were charged for a session that didn’t complete, I’ll flag that for our billing team. You shouldn’t pay for a session that didn’t deliver. Would you like me to note that now?”

Log: session start time, end time, kWh delivered (if available), charge amount, reason for early stop.

Section C — Repeated Failures (Same Driver, Multiple Locations)

If a driver reports that sessions consistently fail to start or end early across multiple chargers on different networks:

1. The fault is almost certainly vehicle-side — specifically the CCS inlet, the vehicle BMS, or the onboard charger.
 2. Do not dispatch a technician to the charging station.
 3. Script: > “Given that you’re seeing this at more than one station, the issue is likely with your vehicle’s charging system rather than the chargers themselves. I’d strongly recommend a visit to your dealership or an EV-certified mechanic. They can run a diagnostic on your charging port and battery management system.”
 4. Log the vehicle make, model, and year for pattern tracking.
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Escalation Criteria

Condition	Action
Hard reset fails, station shows online	Escalate to Tier 2
Same station fails for 3+ drivers in same day	Priority technician dispatch
Vehicle dashboard shows electrical fault	Direct to vehicle manufacturer
Driver was charged for a session that never started	Billing flag + confirm refund timeline
Driver is stranded, no nearby alternatives	Tier 2 + session credit

Agent Reminders

- Always check station status before resetting — resetting an already-faulted unit can corrupt logs.
- If a session has a ghost “in progress” state in the backend, clear it before attempting a new session.
- Never tell a driver their car is broken — say “it may be worth checking the vehicle’s charging system.”
- If the driver was charged for a failed session, flag billing before they hang up — do not make them call back.

KB-002 — Payment & Billing Issues

Category: Payment

Call Volume: ~23% of all support contacts

Trust Impact: HIGH — payment failures are the #1 reason drivers churn from a network

Primary Remote Actions: Session credit flag, billing team escalation, transaction lookup

Why This Matters

Payment issues are disproportionately damaging to driver trust. Among drivers who report a problem, 23% involve payment — and in 19% of those payment cases, the charger accepted payment but the session never started. That means drivers paid for nothing. Even a single such experience can cause a driver to switch networks permanently. Speed of resolution and proactive refund communication are critical.

Background: How Payment Authorization Works

Understanding the payment flow helps agents identify exactly where it broke.

1. **Driver presents payment method** (app wallet, tap credit/debit card, RFID card linked to billing, plug-and-charge certificate)
2. **Pre-authorization hold:** The network places a hold (typically \$25–\$50) on the payment method to confirm it's valid and has sufficient funds
3. **Authorization response:** The payment processor returns APPROVED or DECLINED within a few seconds
4. **Session start signal:** Only on APPROVED does the charger unlock the connector and begin the handshake
5. **Session settles:** At session end (or at billing intervals for long sessions), the actual kWh or time charge is captured; the pre-auth hold is released or adjusted

Failure at steps 2–4 = “charged but no session.”

Failure at step 5 = billing discrepancy.

Section A — Card Declined / Payment Not Going Through

Common Root Causes

Cause	Frequency	Notes
Bank flagged the pre-auth as unusual	~35%	EV charger merchants are often unfamiliar to fraud filters
Card expired or wrong details in app	~20%	Driver may not realize the card on file is outdated
Insufficient funds / credit limit	~15%	Pre-auth amount may exceed available balance
Contactless timeout at reader	~10%	Tap-to-pay fails; chip insert works
Payment processor downtime	~5%	Affects all users on network simultaneously
Corporate card restrictions	~5%	Company cards sometimes block non-travel merchants
	~5%	

Cause	Frequency	Notes
3D Secure / additional authentication required		Card requires bank app confirmation
Virtual card / prepaid card not accepted	~5%	Some networks don't accept prepaid

Diagnostic Script — Card Declined

Step 1 — Identify the payment method

“Are you paying through our app, tapping your card directly on the reader, or using an RFID card?”

If using the app:

“Let me check a couple of things. First, can you open the app and confirm which card is set as your payment method? Sometimes a card expires and the old one stays on file.”

If the card is expired: > “That would explain it — the card on file has expired. You can update it in the app under Payment Settings, and then try starting the session again.”

If the card looks valid: > “Try force-closing the app completely and reopening it, then attempt to start the session again.”

If still failing: > “Everything looks correct on our side, so this is likely a block from your bank — they sometimes flag EV charging transactions as unusual, especially if it's your first time using this network. I'd suggest calling the number on the back of your card and letting them know you're trying to use it at an EV charging station. Alternatively, try a different card.”

If tapping the card directly on the reader:

“Contactless payment can time out on EV chargers because the reader waits longer than a typical point-of-sale terminal. Try inserting your chip instead of tapping — that usually works reliably.”

If RFID card:

“Let me check if your RFID card is active on our system... [look up card status]”

- Card active: Try holding it flat against the reader for a full 3 seconds, not a quick tap.
- Card inactive: Reactivate in the backend and confirm with the driver.
- Card unregistered: Link it to their account. See KB-007.

Step 2 — Check for network-wide payment processor issue

If multiple drivers are reporting payment failures in a short window, check the payment processor status dashboard. If it's a processor outage:

“I can see there’s a temporary issue with our payment system that’s affecting multiple drivers right now — it’s not specific to your card. Our team is working on it. I’m sorry for the timing. Would it help if I noted your details so we can reach out when it’s resolved?”

Step 3 — If all else fails

“I want to make sure you can charge today. A few options: try a different card if you have one, try the app if you were using a physical card, or check if this station accepts QR code or guest checkout. I know this is frustrating and I’m sorry.”

Section B — Charged But Session Never Started

This is the most trust-damaging payment scenario. The driver’s payment was captured or pre-authorized, but the session never delivered energy.

How to Identify

- Driver says they were charged (or saw a hold) but the charger never started
- Session log shows an authorization but no energy delivery
- Driver has a receipt or bank notification but the car is still at the same battery level

Agent Script

“I’m really sorry — you absolutely should not be charged for a session that didn’t run. Let me pull up that transaction right now.”

Step 1 — Locate the transaction

Look up by: account/email, card last 4 digits, approximate timestamp, station ID.

Step 2 — Confirm what the system shows

System shows	Agent action
Pre-auth hold (not settled)	“The charge on your bank statement is a pre-authorization hold — it will automatically drop off within 3 to 5 business days without any action from us. No actual charge was captured.”
Settled charge with \$0 kWh	“You’re right — a charge was captured but no energy was

System shows	Agent action
No transaction found	delivered. I'm flagging this for a full refund right now." "I don't see a matching transaction. Can you share the last 4 digits of the card and the approximate time?"

Step 3 — Flag for refund

"I've flagged this for our billing team. You should see a full refund within 3 to 5 business days. Can I confirm which email address to send your confirmation to?"

Log: station ID, transaction ID, charge amount, timestamp, driver email, session outcome (no energy delivered).

Important: Do not tell the driver a specific refund date beyond the 3–5 business day window. Refund timelines depend on their bank.

Step 4 — Attempt to get them charging today

"While that gets sorted — do you want to try starting a new session? I can try a remote reset on the station first to make sure it's ready."

Section C — Billed Wrong Amount

Driver was charged more than expected

Step 1 — Explain the pricing model

"Happy to walk you through it. At this station, the rate is [per-kWh rate / per-minute rate / flat session fee]. The total for your session would be based on [X kWh delivered / Y minutes]."

Confirm the rate shown matches the posted rate at the station and in the app.

Step 2 — Common discrepancies

Driver complaint	Likely explanation	Script
"I was charged more than the posted rate"	Rate may have changed, or driver used peak pricing period	"Let me check the rate schedule for that station — some locations have different rates at different times of day."
"I was charged for time I wasn't there"	Session may not have been ended properly	"It's possible the session kept running. Always tap End Session in the app or

Driver complaint	Likely explanation	Script
		on the screen — if the cable is unplugged without ending the session, the timer may keep running in some systems.”
“I was charged twice”	Could be pre-auth + final charge appearing simultaneously	“What you may be seeing is a temporary pre-authorization hold and the final charge appearing at the same time. They don’t add up — one will drop off. If both settle, contact us and we’ll refund one immediately.”
“I was charged an idle fee”	Driver left vehicle plugged in after session ended	“Our stations charge an idle fee of [X cents/minute] when a vehicle remains plugged in after the session completes to keep the spot available for other drivers. This is shown in the app and on the station screen before the session starts.”

Step 3 — Billing dispute escalation

“I can’t adjust your bill directly, but I’m flagging this for our billing team with all the details. They’ll review the session data and reach out within [1–2 business days]. Would email or phone be better for you?”

Disputes over \$50 or involving fraud claims go to Tier 2 immediately.

Section D — Pricing Confusion (Pre-Call)

Drivers often call to understand pricing before they charge. This is a service opportunity — a clear, confident explanation prevents a billing dispute call later.

Rate Types and How to Explain Them

Per-kWh pricing (most consumer-friendly, mirrors a utility bill): > “You pay for the actual energy your car receives, measured in kilowatt-hours — the same unit on your electricity bill at home. The rate at this station is [X cents per kWh].”

Per-minute pricing (common on older networks): > “You pay for the time you’re plugged in, not the energy itself. The rate is [X cents per minute]. If your car charges at full speed, this works out to roughly [calculated cost per kWh equivalent] per kilowatt-hour.”

Session fee / flat rate: > “This station charges a flat fee of [X dollars] per session, regardless of how long you charge or how much energy you receive. It’s simple but less cost-effective for short sessions.”

Idle fees: > “Once your session completes, if you leave your car plugged in, an idle fee of [X cents per minute] kicks in. You’ll get a notification in the app before it starts.”

Membership vs. guest pricing: > “If you’re not a member of our network, the guest rate is [X]. Members pay [Y]. Membership costs [Z per month] — if you charge regularly, it usually pays for itself within [estimated sessions].”

Escalation Criteria

Condition	Action
Charged but session never started	Flag for billing review; offer refund timeline
Billing dispute over \$50	Escalate to Tier 2
Driver reports unauthorized charges / fraud	Escalate to Tier 2 immediately; advise driver to contact their bank
Payment processor down (multiple affected)	Notify Tier 2; update status page
Driver threatens chargeback	Do not push back — acknowledge, escalate to Tier 2

Agent Reminders

- Never promise a refund will appear “today” or “immediately” — banks have their own processing timelines.
- Always confirm the driver’s email before closing — the refund confirmation is evidence they’ll need.
- Pre-auth holds look like charges to drivers. Calmly explain the difference before they call their bank.

- An idle fee complaint is almost never a billing error — but acknowledge the frustration before explaining.
- If a driver says they'll dispute the charge with their bank, treat this as urgent — escalate to Tier 2 and log it as a potential chargeback.

KB-003 — Hardware Fault vs. Vehicle-Side Issue Diagnosis

Category: Hardware Diagnosis

Call Volume: ~18% of contacts

Risk: Misattribution wastes technician dispatches and frustrates drivers

Primary Remote Actions: Status check, hard reset, technician dispatch

Why This Matters

Hardware failure is one of the most misdiagnosed categories in EV charging support. Drivers frequently assume the charging station is broken when the fault actually originates in the vehicle — particularly at the charging port, the battery management system (BMS), or the vehicle's onboard communication controller. Dispatching a technician to a healthy charger is expensive and delays resolution for the driver. Equally, dismissing a genuine charger fault as a vehicle problem leaves a broken unit in the field damaging all future users.

This article provides a systematic framework to distinguish charger-side from vehicle-side faults using phone-based questioning.

Background: Where Faults Live

Charger-Side Hardware Faults

Component	What can fail	Typical symptom
Power module / rectifier	Can't convert AC to DC	Session starts, no power flow; screen may show output fault
Connector / cable	Physical damage, bent pins, worn latch	Connector won't lock; intermittent charging
Control board / firmware	Processing failure	Screen frozen, unresponsive, random resets
Payment terminal	Reader failure	Card declined across all cards; screen error
Network modem	No OCPP connectivity	Station shows offline on backend; can't authorize

Component	What can fail	Typical symptom
Ground fault detector	Nuisance tripping or real fault	E005 / GFCI error; do not reset — safety issue
Thermal management system	Overheating protection	Throttles or shuts down in hot weather
Display / UI	Screen failure	Black screen, partial display; may not affect charging

Vehicle-Side Faults

Component	What can fail	Typical symptom
CCS / CHAdeMO inlet	Physical damage, bent pins, debris, corrosion	Won't connect or intermittent loss
Onboard charger (AC)	Internal fault	AC charging fails; DC fine, or vice versa
Battery Management System (BMS)	Communication dropout, calibration error	Session starts then drops; car shows warning
Pilot signal circuit	Can't negotiate power level with EVSE	Handshake fails; error on charger screen
Contactor / relay	Won't close to allow charge	Session authorized but no power flows into battery
Charging limit setting	Soft limit set by driver or manufacturer	Session ends early at a specific SoC

Diagnostic Framework

Phase 1 — Pattern Questions (Ask Before Anything Else)

These five questions narrow the fault location before any remote action is taken.

Q1: Does this happen at only this station, or at other stations too?

- Only this station → likely charger fault. Prioritize reset and tech dispatch.
- Multiple stations → likely vehicle fault. Do not dispatch a tech.
- Only DC fast chargers → narrow to DC inlet or BMS; AC charging may work.
- Only AC chargers → onboard charger suspect; DC may work.

Q2: Have you successfully charged at this same station before?

- Yes, recently → something changed. Could be charger degradation or a vehicle change.

- Never tried here before → no baseline; could be either side.
- Yes, years ago but not recently → charger hardware may have degraded.

Q3: What does the charger screen show?

Screen display	Likely fault location
Blank / black screen	Charger control board or power supply
“Communication error” or “Vehicle not responding”	Vehicle-side (BMS or pilot circuit)
“Connector fault” or “Latch error”	Physical connector issue — could be either
“Ground fault” (E005)	Ground fault — safety issue, do not reset
“Authorized” but no power flow	Could be either; check contactor
“Charging” but car not receiving power	Likely vehicle-side (BMS, contactor)
No error, just stuck on “Starting...”	Backend authorization timeout or network issue

Q4: What does the vehicle display show?

- No warning lights → vehicle presenting as ready; charger fault more likely
- “Charging fault” or specific error code → vehicle-side likely
- “Check charging system” → vehicle-side; recommend dealer visit
- Lightning bolt or plug icon with X → vehicle’s onboard charger may be flagging rejection

Q5: Is the vehicle charging port visibly clean and undamaged?

Ask the driver to inspect: - Any debris, moisture, or foreign objects in the inlet - Bent or damaged pins (they should not touch these) - Discoloration or burn marks (safety flag — escalate) - Corrosion (visible rust or green deposits)

Phase 2 — Remote Action Based on Phase 1

If charger fault is suspected:

1. Check real-time station status in the backend
2. If online: attempt soft reset, then hard reset
3. If offline or faulted: confirm to driver, skip reset, flag for dispatch
4. Log the failure pattern with all Phase 1 answers for the technician

If vehicle fault is suspected:

1. Do not dispatch a technician to the charger — it is unnecessary
2. Script: > “Based on what you’re describing — especially that this is happening at multiple different stations — the issue is most likely with your vehicle’s charging system rather than any specific charger. I know

that's not the answer you were hoping for. I'd recommend contacting your vehicle manufacturer or an EV-certified service center. They can run a diagnostic on your charging port and battery management system."

3. Log: vehicle make, model, year, fault description, charger types attempted (AC/DC), any error codes from vehicle display
 4. Offer to document the call for the driver to share with their dealership
-

Section A — Specific Hardware Fault Scenarios

Scenario 1: Black or Frozen Screen

"Is the rest of the station on — are any lights active, or is the whole unit dark?"

- Whole unit dark: power supply failure or tripped breaker. Hard reset via backend if possible; if no response, dispatch.
- Screen off but charger has indicator lights: display failure only — charger may still function via app or RFID start.
- Screen frozen on startup logo: firmware crash. Hard reset. If persistent: dispatch.

Scenario 2: Charger Shows “Charging” But Car Isn’t Receiving Power

This is a subtle fault — the charger thinks it's working but the vehicle isn't accepting energy.

Checklist: 1. Confirm the vehicle's in-car display shows the charging indicator active 2. Ask: has the state of charge moved in the last 5+ minutes? 3. Ask: is there a charge limit set in the vehicle? (may have already hit the cap)

If car SoC isn't moving and no vehicle-side limit is set: - The contactor (relay that closes to allow current flow) may not be closing, either on the charger side or vehicle side - Hard reset the charger; if unresolved, advise vehicle service

Scenario 3: Error Code “Ground Fault” (E005 or GFCI)

Do not reset this station.

"I need to flag this station as a safety issue — a ground fault error means there's a potential electrical problem at the unit. Please unplug your cable and step away from the station. I'm going to mark it for a technician right now."

Log as a safety-priority dispatch. Do not allow any driver to use this station until the technician clears it.

Scenario 4: Cable / Connector Damage

Ask the driver to look at the connector end (the end that goes into the car) and describe what they see.

- **Bent pins:** Do not attempt to insert — this can damage the vehicle inlet. Dispatch a technician. Offer the driver an alternative station.
- **Burn marks or discoloration:** Safety flag. Remove from service immediately. See KB-009.
- **Debris in socket:** Ask driver to not insert the connector. A technician or the driver (carefully, with the charger unplugged from the car) can remove debris with dry compressed air.
- **Cracked or broken housing:** Remove from service. Dispatch.

Scenario 5: Intermittent Charging (Keeps Starting and Stopping)

This pattern — charging for a few minutes, stopping, restarting — is most commonly caused by:

1. **Loose connector fit** — re-seat the cable; may be a worn latch on the charger or the vehicle inlet
2. **Thermal protection cycling** — charger or vehicle overheating in hot weather; normal behavior but frustrating
3. **Grid instability at the site** — voltage fluctuations trip the charger’s protection; dispatch an electrician via the tech team
4. **BMS communication dropout** — vehicle-side; recommend vehicle service

“That start-stop pattern can happen for a few reasons. Can I ask — is it a hot day, and has the car been sitting in the sun? [If yes] Some chargers and vehicles throttle or cycle in high heat to protect the battery — frustrating, but it’s a safety feature. [If no] Let me try a reset and see if that stabilizes it.”

Section B — Distinguishing Charger vs. Vehicle at Scale

If the same station generates repeated calls from different drivers: - **Fault is at the charger** — dispatch for hardware inspection regardless of remote reset results

If the same driver calls repeatedly across different stations: - **Fault is at the vehicle** — document thoroughly and direct to manufacturer; do not keep dispatching techs

Pattern data to log every call: - Station ID - Vehicle make, model, year - Connector type used (CCS1, CCS2, CHAdeMO, NACS) - AC or DC session - Error code (charger screen + vehicle display) - Whether other stations succeeded or failed for this driver

Technician Dispatch Criteria

Dispatch a technician when:

Condition	Priority
Ground fault error (E005)	SAFETY — immediate
Burn marks or damage to connector	SAFETY — immediate
Hard reset fails, station shows online	High
Connector physically stuck (see KB-008)	High
Same station fails for 3+ different drivers in one day	High
Screen is blank and hard reset has no effect	Medium
Intermittent charging that persists after reset	Medium

Agent Reminders

- Never diagnose a vehicle fault with certainty over the phone — say “it may be worth checking” not “your car is broken.”
- Always log vehicle details (make, model, year, connector type) even when the fault is charger-side — it helps technicians prepare.
- A ground fault (E005) is a hard stop. Do not reset. Do not let the driver retry. Flag and escalate immediately.
- If the driver describes sparks, burning smell, or visible damage: go directly to KB-009 Safety Protocols.

KB-004 — Ghost Availability (App Shows Green, Charger Won’t Start)

Category: Availability / Telemetry

Call Volume: ~8% of contacts

Impact: Disproportionately high — driver drove to a station expecting to charge

Primary Remote Actions: Real-time status check, status flag, nearest-charger lookup

Why This Matters

Ghost availability — where a charging station appears online and available in an app but fails to deliver power — is small in volume but outsized in its harm to driver trust. Drivers make a deliberate trip to the station based on

what the app told them. Arriving to a non-functional charger is one of the most frustrating experiences in the EV ecosystem, especially when time or range is tight. This is also the scenario that predictive fault detection is designed to eliminate upstream.

Background: Why Ghost Availability Happens

Charging station networks rely on OCPP (Open Charge Point Protocol) messages to track the state of each unit. A charger is marked “available” when its last OCPP heartbeat and status notification said it was. The problem: several failure modes can break the charger’s ability to charge without affecting its ability to send heartbeat pings.

Root Causes

Cause	What’s happening	Why app still shows green
Stale telemetry / lost OCPP session	Charger went offline but network didn’t receive a clean “offline” message	Last status was “Available”; system assumes that’s still true
Payment terminal fault	Card reader is broken; charger is online but can’t authorize sessions	Charger pings network fine; payment module is separate
Firmware hang in authorization module	Charger responds to heartbeat but authorization requests time out	Health check doesn’t test authorization path
Network connectivity intermittent	Charger SIM or Ethernet drops; heartbeats get through, session commands don’t	Status updates arrive, command responses don’t
Power module failure	Charger electronics are on (screen, modem) but power conversion hardware is dead	Control systems operational; power stage failed silently
OCPP version mismatch	A backend update changed the command format; charger can’t process new start-session messages	Heartbeats are version-agnostic; session commands aren’t
Cable/connector	Physical damage not yet triggering a fault status	Self-test hasn’t caught the physical failure yet

Cause	What's happening	Why app still shows green
undetected fault		

Diagnostic Script

Step 1 — Confirm the scenario

“I completely understand — it’s incredibly frustrating to drive to a station that the app showed as available. Let me check its actual real-time status on our end right now.”

Pull real-time station status from the backend operations console (not the driver-facing app status — that may be cached).

Step 2 — Interpret what the backend shows

Backend status	Meaning	Agent action
Available (same as app)	Telemetry lag — charger may be in a failure state we haven’t detected yet	Attempt soft reset; if fails, flag as ghost unit
Faulted	Charger has reported a fault, but app cached the last “Available” state	Confirm to driver, apologize, offer nearest alternative
Offline	Charger lost connectivity — OCPP heartbeat stopped	Confirm to driver, flag for connectivity investigation
In session (but driver says no one is charging)	Ghost session — an orphaned session is blocking new starts	Clear the ghost session from the backend, then retry
Reserved	Station was reserved by another driver via app	Explain reservation system, offer nearest unreserved unit

Step 3 — Script for confirmed ghost (backend shows available but charger unresponsive)

“You’re right — even though the app showed it as available, this station isn’t actually responding to our systems right now. I’m flagging it so it shows correctly for other drivers and gets looked at by our team.”

Mark the station with status: “Ghost — availability mismatch.” This triggers a tech inspection and a status update push to the app.

Step 4 — Clear a ghost session (if backend shows “in session”)

“It looks like there’s a stuck session from a previous driver that’s preventing new ones from starting. I’m clearing that now — it’ll take about 30 seconds.”

Clear the orphaned session in the backend. Attempt a soft reset. Ask driver to retry.

Step 5 — Soft reset attempt (for ghost units showing “available”)

“I’m going to try a remote restart on the station and see if that brings it back. It’ll take about 30 seconds.”

Trigger soft reset. If the charger responds and comes back online clean — ask driver to retry starting a session.

If soft reset doesn’t restore function:

“The station isn’t coming back online through a restart. This will need a technician to look at. I’m really sorry you came out here for this.”

Step 6 — Offer real-time-verified alternative

Do not simply read off the next nearest station from the map — that map may have the same staleness problem.

“Before you head anywhere else, let me verify the status of the nearest alternatives on our end — not just the app, which can lag. Give me just a moment.”

Check backend real-time status for the 2–3 nearest stations. Confirm at least one is genuinely live before recommending.

“I can confirm [Station Name / Address] is showing as live and available on our side right now — not just the app. That’s [X miles] from where you are.”

Section A — Explaining the Problem to Drivers Without Jargon

Drivers want to know why the app lied to them. A brief, honest explanation builds more trust than deflection.

Script: > “What happened here is a timing gap between what the station last told our network and what’s actually happening right now. Normally that gap is just a few seconds, but if the station loses its connection in a specific

way, the last status it sent — ‘available’ — can stick around longer than it should. We’re working on making that detection faster, and I’ve flagged this one so it updates correctly.”

Avoid saying: “the app is wrong” (implies it’s always unreliable), “that’s just how it works” (dismissive), or “you should have called first” (blaming the driver).

Section B — Ghost Session Clearing

A ghost session is an orphaned OCPP session — a session that was opened (possibly by a previous driver’s failed attempt, a firmware crash, or a connectivity loss mid-session) that was never cleanly closed.

Effects: - Station shows “In Session” or “Occupied” to new drivers - New session commands are rejected by the charger - Energy meter may be running on the ghost session, which can generate erroneous charges

To clear: 1. Identify the session ID in the backend 2. Send a remote stop command to that session ID 3. Wait 30 seconds for the charger to confirm the stop 4. Verify the station returns to “Available” in the backend 5. Attempt a soft reset to clear any residual state 6. Ask the driver to try again

If the ghost session has a billing record associated with it, flag it for billing review — it may have generated a charge against a previous driver’s account.

Section C — High-Frequency Ghost Stations

Some stations ghost-out repeatedly — the same unit generates multiple availability mismatch flags per week. These are candidates for:

1. **Firmware upgrade** — a known bug in certain firmware versions causes intermittent heartbeat continuation without functional authorization
2. **SIM / modem replacement** — if the connectivity is intermittent at the hardware level
3. **OCPP configuration review** — incorrect heartbeat interval or status notification settings
4. **Physical inspection** — a power module fault that hasn’t yet triggered a full fault status

Log every ghost-availability call with station ID. If the same station generates 3+ ghost flags in a 7-day period, escalate to Tier 2 for engineering review rather than repeated tech dispatches.

Driver Impact Mitigation

When a driver has made a trip to a non-functional station, consider proactively:

1. **Session credit / goodwill credit** — flag for billing team if the driver was inconvenienced significantly
2. **Status update subscription** — some apps allow drivers to subscribe to status alerts for a specific station; suggest this if available
3. **Feedback note** — log the driver’s experience as explicit product feedback; ghost availability is a known systemic issue that engineering teams should be aware of

Escalation Criteria

Condition	Action
Ghost confirmed, soft reset fails	Flag for tech inspection; offer real-time-verified alternative
Ghost session cleared, but station goes ghost again within the hour	Escalate to Tier 2 engineering
Same station ghosts 3+ times in 7 days	Escalate to Tier 2; remove from recommended alternatives until cleared
Driver drove significant distance and is now stranded	Session credit + Tier 2 escalation
Ghost session had billing implications for another driver	Billing flag for that driver’s account

Agent Reminders

- Never recommend a nearest-alternative station without verifying its backend status first. The app map can be stale.
- Always flag ghost stations — a ghost unit that isn’t flagged keeps fooling other drivers.
- If a ghost session exists on the backend, clear it before doing a reset — the reset alone may not clear an orphaned session record.
- Ghost availability is a known industry-wide problem; acknowledge it honestly to drivers rather than deflecting.

KB-005 — CCS Charging — Full Driver Guide & Troubleshooting

Category: CCS Connector — Education & Troubleshooting
Call Volume: ~12% of contacts
Audience: Both new EV drivers unfamiliar with CCS and experienced drivers hitting CCS-specific faults
Primary Remote Actions: Remote reset, status check, technician dispatch

Why This Matters

CCS (Combined Charging System) is the dominant fast-charging standard in North America and Europe, used by most non-Tesla EVs. However, its two-part connector design, handshake complexity, and physical locking mechanism generate confusion and specific failure modes not seen with AC charging. Agents must be fluent in both explaining CCS to new users and diagnosing CCS-specific technical failures.

Background: What Is CCS?

CCS Standards

Standard	Also called	Used in	Connector description
CCS1 (CCS Type 1)	SAE J1772 Combo	North America	J1772 AC pins on top + 2 large DC pins on bottom
CCS2 (CCS Type 2)	IEC 62196 Combo	Europe, some international	Mennekes AC pins on top + 2 large DC pins on bottom

The top portion (AC pins) handles Level 2 AC charging. The bottom portion (DC pins) handles DC fast charging (DCFC). Both are part of a single combined connector — hence “Combined Charging System.”

Power Levels

Level	Power range	Approximate charge speed
Level 2 AC (CCS AC portion)	7–22 kW	Adds ~20–60 miles per hour
DC Fast Charging	50–350 kW	Adds ~100–300+ miles per 20–30 min

Level	Power range	Approximate charge speed
(CCS DC portion)		

How to Identify a CCS Charger

- DC Fast Charger with a large, two-tier connector (top half smaller, bottom half with 2 large round pins)
- Often labeled “CCS,” “DC Fast,” or “DCFC” on the station
- Typically tethered (cable attached to charger, not a loose socket)
- Usually has a large rectangular display showing session status

Which Vehicles Use CCS

Most non-Tesla EVs sold in North America and Europe use CCS1 (NA) or CCS2 (EU), including: - BMW, Chevrolet, Ford, Hyundai, Kia, Nissan (newer models), Polestar, Rivian, Stellantis brands (Jeep, RAM), Volkswagen Group (Audi, Porsche, VW)

Note: Tesla vehicles use NACS (North American Charging Standard) natively, though many newer Teslas include a CCS adapter or NACS-to-CCS compatibility. CHAdeMO (used by older Nissan LEAF and Mitsubishi) is a separate standard entirely and is not CCS.

Section A — How to Start a CCS Charging Session (Driver Education)

Many calls come from drivers who are new to CCS and simply don’t know the correct process. Walk them through this step by step.

Step-by-Step CCS Session Start

Step 1 — Check vehicle compatibility > “First — does your car have a CCS port? It’s on the outside of the vehicle and looks like a standard J1772 socket on top with two larger pins below it. If you’re not sure, check your owner’s manual or look up your vehicle model.”

Step 2 — Pull up to the charger Park close enough that the tethered cable reaches the vehicle inlet without strain. Overstretched cables can damage the connector and the inlet.

Step 3 — Power down the vehicle (if required) Most modern EVs allow charging while the car is powered on, but some manufacturers require the vehicle to be in Park and powered down (key off or in ready mode). Check vehicle-specific guidance: - BMW, Hyundai, Kia: typically fine in “ready” mode - Some older EVs: may require powered down

Step 4 — Locate the charging port on the vehicle The CCS inlet is typically on the driver's side rear quarter panel, front quarter panel, or rear center (varies by manufacturer). Open the port cover — this may be a mechanical flap, electronic button, or automatic (opens when a compatible connector is detected).

Step 5 — Prepare the connector Pick up the CCS connector. It is heavy compared to an AC plug. Hold it by the handle — not the cable. Align the connector with the inlet, paying attention to orientation (the connector is keyed and will only fit one way).

Step 6 — Insert the connector Insert firmly until you feel and/or hear a click. The latch should engage. Do not force it. If it doesn't click: - Check alignment — the connector may be slightly rotated - Inspect for debris in the vehicle inlet - Try a slightly different angle

Step 7 — Authorize the session Authorization methods vary by station: - **App:** Open the network app, tap Start Session, select the station - **RFID card:** Tap card against the RFID reader on the charger (usually marked with a contactless payment icon) - **Tap-to-pay:** Tap credit or debit card against the card reader - **QR code:** Scan the QR code on the station with your phone to open a guest checkout flow - **Plug-and-charge (ISO 15118):** Some vehicles and networks support automatic authentication when the cable is plugged in — no additional action needed

Step 8 — Confirm session started - The charger screen should show "Charging" or a session in progress - The vehicle's dashboard or instrument cluster should show a charging indicator - You may hear a relay click as the contactors close and power begins flowing - The connector latch will lock — you will not be able to remove the cable while the session is active

Step 9 — During the session - The vehicle can be locked (and should be, for security) - You can monitor progress in the app or on the charger screen - Do not kick, pull, or force the cable — it is mechanically locked

Step 10 — Ending the session - Tap "Stop Session" in the app OR on the charger screen - Wait for the screen to confirm the session is complete - The connector latch will release - Remove the connector and return it to the holder - Close the vehicle port cover

Section B — CCS-Specific Troubleshooting

Problem: Connector Won't Insert

“Can you describe what's happening — is it not going in at all, or going in partway?”

Not going in at all: 1. Alignment issue — rotate slightly and re-try. The connector is keyed. 2. Debris in vehicle inlet — look inside with a flashlight. Do not insert the connector if debris is present. 3. Wrong connector type —

confirm the vehicle uses CCS. CHAdeMO and NACS connectors look different. 4. Physical damage to connector — if pins are visibly bent, do not insert. Dispatch a technician.

Going in partway but not clicking: 1. The vehicle inlet cover may not be fully open — check that the port door is completely retracted. 2. The connector may need a firmer push — CCS connectors require more insertion force than AC plugs. 3. Inlet damage — if the vehicle's CCS port has debris, corrosion, or damage, do not force it.

Problem: Session Won't Start After Connector Is Inserted

Follow the diagnostic in KB-001 (Session Won't Start), with these CCS-specific additions:

CCS handshake failure: The CCS DC charging handshake (ISO 15118 / DIN 70121 / IEC 61851) is more complex than AC charging. It involves: - Control Pilot (CP) signal negotiation - Proximity Pilot (PP) confirmation that the connector is properly seated - High-level communication over PLC (Power Line Communication) for ISO 15118

If the handshake fails: 1. Try unplugging and reinserting firmly — PP signal depends on a complete mechanical connection 2. Soft reset the charger 3. Ask if the driver has tried a different CCS charger recently — if all CCS fast chargers fail, the vehicle's PLC controller may have a fault

Charger screen shows “Vehicle not responding” or “Communication error”: Vehicle-side pilot signal or PLC communication fault. See KB-003. This is vehicle-side.

Charger screen shows “Connector not detected”: The PP (Proximity Pilot) circuit isn't registering a seated connector. 1. Unplug and re-seat firmly 2. Check connector and inlet for debris 3. If the connector latch mechanism is worn, it may not be generating the PP signal — dispatch a technician

Problem: DC Fast Charging Won't Start but Level 2 (AC) Works

This pattern — DC fails, AC succeeds at the same vehicle inlet — points to: 1. **Vehicle DC inlet fault** — the DC pins (lower portion of CCS) may be damaged while AC pins (upper) are fine 2. **Vehicle BMS or DC fast charge module fault** — the vehicle's DC fast charge pathway has a software or hardware issue 3. **This is almost always vehicle-side**

“If your car charges fine on a regular Level 2 charger but DC fast charging consistently fails, the issue is with your car's DC fast charging system rather than the stations. I'd recommend a visit to your dealership — they can run a specific diagnostic on the DC charging pathway.”

Problem: Charging Stops Mid-Session on CCS

Refer to KB-001 Section B (Session Ended Early), with these CCS-specific notes:

Thermal derating on DC fast charging: DC fast charging generates significant heat in the vehicle battery, the charger power module, and the cable itself. Thermal management systems on both sides will reduce power or stop the session if temperatures exceed safe limits. This is normal and expected, especially: - In hot weather (ambient above 30°C / 86°F) - After a previous fast charge session (battery already warm) - At high SoC (above 80%)

Communication dropout mid-session: If the PLC communication link drops during a session (due to charger modem issues, electrical interference, or a loose connector), the charger will stop power delivery as a safety measure. 1. Re-seat the connector 2. Soft reset and retry

Vehicle charge limit reached: Many EVs are factory-set to a charging limit (often 80% or 90%) to protect battery longevity. The driver may not be aware.

“It’s worth checking your vehicle’s charging settings — most EVs let you set a maximum charge level. If it’s set to 80%, the session will end at 80% even if you expected it to go higher.”

Problem: Connector Stuck in Vehicle

This is covered fully in KB-008 (Stuck Cable), but CCS-specific points:

- The CCS connector locks mechanically when a session is active. It will not release until the session officially ends.
- If the session has ended on the screen but the connector is still locked: send a remote connector unlock command.
- Some vehicle models have a manual emergency cable release inside the vehicle (typically in the trunk or under a panel). Check vehicle documentation.
- Do not advise the driver to pull forcefully — this can damage the vehicle’s CCS inlet, which is an expensive repair.

Section C — CCS vs. Other Connector Standards

Agents must be able to quickly identify if a driver has the wrong connector type, as this is a common source of confusion.

Standard	Connector look	Common vehicles	Compatible with CCS charger?
CCS1		Most US non-Tesla EVs	Yes — this is CCS

Standard	Connector look	Common vehicles	Compatible with CCS charger?
	J1772 top + 2 DC pins bottom		
CCS2	Mennekes top + 2 DC pins bottom	EU EVs	Yes — this is CCS (EU variant)
CHAdMO	Round, single-piece DC connector, separate from AC	Older Nissan LEAF, Mitsubishi	No — needs CHAdMO port
NACS (Tesla)	Slim, single-piece oval connector	Tesla, some new Ford/GM/Rivian	Adapter required for CCS
J1772 only	Standard AC plug, no DC pins	All EVs (AC charging only)	Only for Level 2, not DCFC
GB/T	Chinese standard	Chinese-market EVs	No

If a driver with a CHAdMO vehicle is at a CCS-only fast charger: > “Unfortunately, CHAdMO and CCS are different connector standards and aren’t compatible with each other. That station only has CCS ports. Do you want me to help you find the nearest station that has a CHAdMO connector?”

If a Tesla driver is at a CCS station: > “Tesla’s native connector is NACS. Some CCS stations have a NACS port or a CCS-to-NACS adapter available — does this station have one? If not, you’d need an adapter or a Tesla Supercharger.”

Section D — CCS Station Identification Guide (For Agents)

When a driver can’t find the station ID, agents can help identify the charger type from the driver’s description:

Driver describes	Likely charger type
“Big box on a pole, thick cable, two big pins at the bottom”	DC Fast Charger with CCS
“Small box on a wall, looks like a gas pump nozzle”	Level 2 AC charger (J1772)
“Round connector, single piece”	CHAdMO

Driver describes	Likely charger type
“Slim, one-piece connector, Tesla logo”	Tesla Supercharger / NACS
“Two connectors hanging off the same unit”	Dual-port DCFC (often CCS + CHAdeMO)

Escalation Criteria

Condition	Action
DC fails, AC works (same inlet)	Vehicle-side — direct to manufacturer
CCS connector has bent or damaged pins	Dispatch technician; do not allow use
Connector locked and won’t release after session end	Remote unlock; if fails, dispatch
Repeated CCS handshake failures across multiple stations	Vehicle BMS/PLC issue — direct to dealer
Driver reports sparks or burning from CCS connector	KB-009 Safety Protocols immediately

Agent Reminders

- CCS connectors are heavier and require more insertion force than AC plugs — this surprises new users.
- If DC fast charging fails but AC works, it is almost always the vehicle, not the charger.
- The connector locks while charging. Trying to remove it during a session is normal driver behavior — calmly explain they need to end the session first.
- CHAdeMO and CCS are not compatible. Know the difference and have a plan for CHAdeMO drivers at CCS-only stations.
- “Plug-and-charge” (ISO 15118) is increasingly common — some vehicles start the session automatically when plugged in. If a driver says “I don’t know how to start it,” ask if their car does plug-and-charge first.

KB-006 — Slow or Throttled Charging

Category: Charging Performance

Call Volume: ~6% of contacts

Resolution Type: Mostly educational; some hardware resets

Primary Remote Actions: Soft reset, power output check

Why This Matters

Slow charging complaints are among the most nuanced in EV support. Most of the time, the charger is performing correctly and the reduced speed has a legitimate technical explanation — battery state of charge, thermal management, power sharing, or a vehicle charge limit. However, a small portion of slow-charging complaints reflect genuine hardware faults, such as a degraded power module or a throttled rectifier. Agents must quickly distinguish “expected behavior” from “actual fault” to avoid unnecessary tech dispatches while also not dismissing drivers experiencing real problems.

Background: Why EV Charging Speed Varies

Unlike filling a gas tank, EV charging speed is not constant. Multiple factors dynamically adjust how fast energy flows into the battery:

1. Battery State of Charge (SoC) Curve

DC fast charging follows a charging curve: - **0–20% SoC:** Charger can deliver near full rated power - **20–80% SoC:** Sustained high power (this is the “fast” part of fast charging) - **80–100% SoC:** Power tapers progressively to protect battery chemistry

This taper at high SoC is engineered behavior — it’s not a fault. A driver who joins a charger at 85% will always see slow speeds.

Script: > “DC fast chargers automatically slow down once your battery gets above about 80% — that’s a built-in protection for your battery’s long-term health. It’s completely normal and actually a good sign that your battery management system is working correctly.”

2. Thermal Management (Battery and Charger)

Both the vehicle battery and the charger's power modules generate heat during fast charging. If temperatures exceed safe thresholds: - The vehicle BMS will reduce the accepted charge rate - The charger's thermal protection will reduce output - In extreme cases, charging may pause briefly to allow cooling

This is most common in: - Hot ambient conditions (above 30°C / 86°F) - Cold ambient conditions (below 5°C / 41°F) — batteries also charge slowly when cold - Sessions following a previous DC fast charge (battery already warm) - Vehicles with less effective thermal management systems

Script: > “In hot weather, both your car's battery and the charger itself slow down to manage heat. It's a safety feature, not a malfunction — the speed should pick up if temperatures drop or if you wait a bit between charging sessions.”

3. Power Sharing Between Adjacent Vehicles

Many DC fast charging stations share a total power budget across multiple connector pairs. If two vehicles are charging simultaneously: - The total available power is divided between both sessions - The split may be equal or weighted (by vehicle capability, SoC, or arrival order, depending on firmware) - Adding a second vehicle to an adjacent port can immediately halve the power to both

Script: > “I can see there's another car charging at the adjacent port on that same unit. Some stations share power between vehicles plugged into the same unit. Moving to a different station — not just a different port on the same unit — would get you the full available speed.”

4. Vehicle Maximum Accepted Rate

Every EV has a maximum DC fast charging rate set by its hardware and BMS. Plugging a vehicle into a 350 kW charger doesn't mean the vehicle will accept 350 kW — it accepts only what it can handle:

Example vehicle	Max DC fast charge rate
Nissan LEAF (24–40 kWh)	50 kW
Nissan LEAF Plus (62 kWh)	100 kW
Hyundai IONIQ 5	230 kW
Kia EV6 GT	240 kW
Ford Mustang Mach-E	150 kW
Chevrolet Bolt EV	55 kW
Rivian R1T/R1S	220 kW

If the driver expects 350 kW but the vehicle can only accept 50 kW, the station is performing correctly.

Script: > “What’s the maximum fast charging speed for your vehicle? Some EVs are designed to accept a lower rate — the charger will always match what your car can handle, so the station speed rating is a ceiling, not a guarantee.”

5. Vehicle Charge Limit Set

Many drivers set a charge limit in their vehicle app or settings to extend battery longevity. A limit of 80% means the car will slow to nearly zero as it approaches 80% and stop entirely when reached.

Script: > “It’s worth checking your vehicle’s app or charging settings — a lot of drivers set a charge limit at 80 or 90 percent to protect their battery. If there’s a cap set, the charger naturally slows as you approach it.”

6. Grid Voltage Issues at the Site

If the electrical supply to the charging site has voltage irregularities or is undersized for the installed charger capacity, the charger’s power output will be limited. This is a genuine fault condition — neither the vehicle nor the charger firmware is throttling, but power input is insufficient.

Signs of a grid issue: - Consistently low output at the same station across multiple vehicles - Station shows “power limited” or a reduced output indicator - Output drops further when multiple chargers on site are in use simultaneously

This requires investigation by an electrician, not a remote reset.

Diagnostic Script — Slow Charging

Step 1 — Get the baseline numbers

“What speed are you currently seeing — does the screen or your vehicle’s display show a kW number or a miles-per-hour estimate?”

“And what speed were you expecting at this station?”

“What’s your current battery percentage?”

Step 2 — Apply the decision tree

Battery above 80%: → Expected taper. Explain SoC curve. No fault.

Cold or hot weather conditions: → Expected thermal management. Explain. No fault unless dramatically below expected (e.g., 10 kW from a 150 kW rated charger in mild weather).

Two vehicles at the same station unit: → Power sharing. Explain. Suggest moving to a different unit.

Vehicle's max charge rate is lower than the station rating: → Expected. Explain vehicle rate ceiling. No fault.

Driver has a charge limit set: → Expected. Explain limit. No fault — driver can adjust in vehicle settings.

None of the above apply AND the power is dramatically low: → Potential genuine fault. Proceed to Step 3.

Step 3 — Check backend power output

Check the station's reported output in the backend: - If reported output matches what the driver sees → the charger is reporting accurately; fault may be on vehicle side - If reported output is significantly higher than what the driver reports receiving → measurement discrepancy; log for investigation - If the charger shows "power limited" or a degraded output state in the backend → genuine charger fault

Step 4 — Soft reset for genuine fault

"What you're describing does sound like more than the usual throttling. I'm going to send a reset to the station — sometimes that restores the full output."

Trigger soft reset. Ask driver to check the power level after the session restarts.

If power is restored: session was in a degraded state, likely a firmware or module issue that reset cleared. Log for monitoring.

If power is still dramatically low after reset: genuine hardware degradation or fault.

"The station still isn't delivering its full rated power. I'm going to flag it for a technician to inspect the power modules. I'm really sorry about this."

Section A — Specific Slow-Charging Scenarios

Scenario: L2 Speed on a DCFC Port

Driver is at a DC fast charger but receiving Level 2 speeds (roughly 7–22 kW instead of 50–350 kW).

Most common causes: 1. **DC portion of CCS not connected properly** — only AC pins are making contact. Ask driver to unplug and re-seat firmly. 2. **Vehicle defaults to AC charging** — some vehicles require DC fast charge to be enabled in settings. Check vehicle documentation. 3. **Charger power**

module for DC has faulted — the AC portion still works, DC portion is dead. This is a charger hardware fault. 4. **OCPP backend sent an AC-only session start** — a backend configuration error. Soft reset and retry.

Scenario: Charges Fast Initially Then Drops to Nearly Zero

Classic SoC taper at high state of charge. If this is happening below 80%, it may be: - Battery thermal derating (cell temperatures too high) - BMS conservatism (some manufacturers are more aggressive than others) - A degraded cell group in the battery pack causing BMS to limit overall charge rate

This is vehicle-specific and generally not actionable by charging network support.

“That slowdown near the top of the battery is normal for most EVs — the battery management system protects the cells by tapering the charge rate. If it’s dropping dramatically at a lower percentage than expected, it may be worth mentioning to your vehicle dealer, but the charger is performing correctly.”

Scenario: Power Fluctuates Up and Down During Session

Intermittent power that oscillates suggests: 1. Grid instability at the site (genuine electrical infrastructure issue — dispatch) 2. Vehicle BMS negotiating and renegotiating the charge rate (may be normal for some vehicles) 3. Connector partially seated — causing intermittent contact (ask driver to re-seat) 4. Thermal cycling — power dips briefly while cooling, then resumes (expected in hot conditions)

Scenario: Consistently Slow at This Station vs. Others

Driver reports same vehicle charges at full speed at other stations but always slow at this specific one:

This is almost certainly the charger’s power module operating at reduced capacity. Log the station ID, the reported speed, and the driver’s vehicle. Flag for a technician power output test — many network operators have a power output verification tool in the backend.

Escalation Criteria

Condition	Action
Power dramatically below station rating, no thermal/SoC explanation	Flag for tech inspection
L2 speeds at a DCFC port, re-seat and reset didn’t help	Hardware fault dispatch
	Priority tech inspection

Condition	Action
Same station slow for multiple different vehicles	
Driver reports power fluctuating wildly	Grid issue — escalate to tech (electrical)

Agent Reminders

- Never tell a driver the slow speed is “definitely normal” without walking through the decision tree — some genuine faults present like expected throttling.
- The SoC taper is the most common explanation — if the driver is at 80%+, lead with that.
- Cold batteries are a real limitation. Below 10°C / 50°F, DC fast charging can be significantly reduced — especially on vehicles without active battery preconditioning.
- Power sharing is often invisible to the driver — they don’t see the other vehicle. Check the backend for simultaneous sessions on the same unit.
- If the charger shows full output but the vehicle reports slow charging, the issue is vehicle-side. Do not dispatch a tech to the charger.

KB-007 — App, Account & RFID Card Issues

Category: Authentication & Account

Call Volume: ~9% of contacts

Resolution Rate on Call: ~85%

Primary Remote Actions: Password reset, RFID card activation, account lookup

Why This Matters

App and account issues gate everything else — a driver who can’t authenticate can’t start a session regardless of charger health. These issues are often quick to resolve but require careful verification to avoid account security risks. RFID card issues are especially common among fleet drivers and corporate accounts who use physical cards rather than the app.

Background: Authentication Methods

Most EV charging networks support multiple ways to authenticate and start a session:

Method	How it works	Common issues
Network app	Driver opens the app, selects station, taps Start	Login failures, app crashes, stale session
RFID card	Tap physical card against reader	Card not registered, card inactive, reader fault
Tap-to-pay	Tap contactless credit/debit against payment reader	Timeout, bank block, unsupported card
QR code / guest checkout	Scan QR on charger, pay as guest	Varies by station; no account required
Plug-and-charge (ISO 15118)	Vehicle auto-authenticates when cable inserted	Vehicle must support; requires pre-enrollment
Phone number / SMS code	Some networks send a one-time code via SMS	Delivery delay, wrong number on file

Section A — App Issues

Problem: Driver Can’t Log In

Step 1 — Identify the failure type

“What happens when you try to log in — do you get an error message, or does it just not respond?”

Symptom	Likely cause
“Invalid password”	Wrong password; needs reset
“Account not found”	Wrong email address; or account doesn’t exist
“Account locked”	Too many failed attempts
App crashes on login	App bug; try update or reinstall
Loads but says “session expired”	Token expired; force logout and back in
Email/password correct but 2FA isn’t coming	SMS delivery issue or wrong number on file

Step 2 — Password reset

“Let me send a password reset to your email. What address do you have on file with us?”

Look up the account to confirm the email is active before sending the reset.

“I’ve sent a reset link — it should arrive within a minute or two. If you don’t see it, check your spam folder. The link expires in 24 hours.”

If the driver doesn’t have access to the email on file: > “I can update your email address, but I’ll need to verify your identity first. Can you confirm your name and the last 4 digits of the card on file with us?”

After identity verification, update the email and resend the reset.

Step 3 — Account locked

If the account is locked due to too many failed login attempts: > “Your account has been temporarily locked for security. I can unlock it right now — give me just a moment.”

Unlock the account in the backend. Then: > “I’ve unlocked it. Try logging in again, and if you’re not sure of your password, let me send a reset link too.”

Step 4 — Account not found

“I’m not finding an account with that email. Is it possible you signed up with a different address, or through a third-party like Google or Apple sign-in?”

Check for: - Common email typos (gamil vs. gmail, .con vs. .com) - Social login (Google, Apple, Facebook) — the account may exist but be linked to a social provider, not a password - A different email format (firstname.lastname vs. first_last, etc.)

If genuinely no account exists: > “It doesn’t look like there’s an account with that email. You can sign up in the app in about a minute — or if you need to charge right now, most of our stations accept a credit card tap directly, no account needed.”

Problem: App Won’t Start a Session (Account OK, Login Fine)

Step 1 — Verify the app version Outdated apps can fail to communicate with updated backends.

“Quick check — when did you last update the app? It’s worth checking if there’s an update available in the App Store or Google Play.”

Step 2 — Force close and reopen

“Try fully closing the app — don’t just minimize it — and reopening it. Then try starting the session again.”

Step 3 — Check location permissions

Many network apps require location access to confirm the driver is near the station before authorizing a session.

“Can you check that the app has permission to access your location? That’s sometimes required to start a session. You can find it in your phone’s Settings under the app name.”

Step 4 — Clear app cache (Android) or reinstall (iOS/Android)

If none of the above work:

“Try uninstalling the app and reinstalling it fresh. That clears any corrupted data that might be blocking it. Your account and payment info are stored on our servers, so you won’t lose anything.”

Step 5 — Use alternative method

“While we sort this out — you can also start the session by tapping your credit card directly on the station reader. Would you like to try that?”

Problem: App Shows Wrong Charger Status / Map Not Updating

Stale app data — see KB-004 for full context. Short version:

“The app’s map can sometimes lag behind real-time status. Let me check the actual live status on our end and tell you what it’s really showing.”

Advise driver to pull-to-refresh or toggle location off/on to force a map refresh. If the app consistently shows wrong data, clearing the cache or reinstalling helps.

Section B — RFID Card Issues

Problem: RFID Card Won’t Work at the Reader

Step 1 — Check card status in backend

Look up the card number (usually printed on the card or the card packaging).

Backend status	Action
Active, registered to account	

Backend status	Action
	Likely a tap technique issue or reader fault
Inactive	Activate the card
Not found / unregistered	Register the card to the driver's account
Suspended	Cannot reactivate without Tier 2 review — escalate
Expired	Issue a replacement card

Step 2 — Tap technique (if card is active)

Most RFID card failures are technique issues:

“Try holding the card flat against the reader — not at an angle — for a full 3 to 5 seconds. It should beep or flash when it reads. A quick tap sometimes isn't long enough.”

Also: - Ensure no other RFID cards (bank cards, access badges) are stacked behind the charging card — they interfere with each other - The card should be presented to the specific reader zone, usually marked with a contactless payment icon or RFID wave symbol - Some readers require the card to be within 2–3 cm (about 1 inch)

Step 3 — Reader fault

If the card is confirmed active and tap technique is correct, the reader may have a hardware fault.

“Your card looks active and correct on our end. If it's still not reading, the RFID reader on that specific unit may have a fault. You can try starting via the app or tapping a credit card directly as an alternative, and I'll flag the reader for inspection.”

Problem: RFID Card Is Not Registered

Step 1 — Verify driver identity

Before linking a card to an account, verify: > “Can you confirm the email address on your account and the last 4 digits of the payment card on file?”

Step 2 — Register the card

Enter the RFID card number (printed on the card, usually 16–20 digits) into the account in the backend.

Step 3 — Confirm registration

“I've linked your RFID card to your account. Give it a try — tap it against the reader now.”

Problem: Fleet / Corporate RFID Account

Corporate accounts sometimes have different authorization flows: - Cards may be assigned to a specific vehicle, not a specific driver - Cards may have charging time restrictions or location restrictions - Cards may require a PIN - Cards may be managed by a fleet administrator who needs to activate or authorize them

“Your card is registered to a corporate fleet account. Changes to fleet cards typically need to go through your company’s fleet administrator — I can’t activate or modify fleet cards directly. I’d recommend contacting your fleet manager. As an alternative, you can use a personal payment method to charge today.”

Section C — Plug-and-Charge Setup Issues

Plug-and-charge (ISO 15118 protocol) automatically authenticates the session when the cable is inserted, without any app or card interaction. It requires: 1. The vehicle to support ISO 15118 (check manufacturer documentation) 2. The charging station to support ISO 15118 3. The driver to have enrolled their vehicle’s charging certificate with the network

Common issues:

Issue	Cause	Resolution
Plug-and-charge doesn’t start automatically	Vehicle not enrolled, or network not configured	Enroll vehicle via app settings, or use app/RFID for this session
Plug-and-charge authorizes to wrong account	Multiple accounts or vehicles configured	Review plug-and-charge settings in app
Plug-and-charge worked before, not now	Certificate may have expired	Re-enroll in app settings
Driver wants to disable plug-and-charge	Preference	Direct to app account settings

“Plug-and-charge requires a bit of setup — your vehicle’s certificate needs to be enrolled with our network. You can do that in the app under Account > Vehicles > Plug-and-Charge. For today’s session, try the app or card as a backup.”

Section D — Account Management on Call

Updating Payment Method

Do not update payment details over the phone — direct drivers to the app or web portal for PCI compliance.

“For your security, I can’t update card details over the phone. You can update your payment method in the app under Account > Payment — it only takes a minute.”

Account Closure

Treat account closure requests with care. Confirm identity first, then:

“I can close your account. Before I do, can I ask what prompted this? If there’s a specific issue, I’d like to see if we can resolve it. If you’d still like to close it, I’ll need to confirm your identity.”

Process: verify name, email, confirm any outstanding balance is settled, then initiate closure per standard process.

Transferring an Account to a New Email

Verify identity (name + last 4 of card on file), then update email in backend. Log the change.

Escalation Criteria

Condition	Action
Driver locked out and can’t access email on file	Tier 2 identity verification
Suspected account compromise (unauthorized charges, unknown sessions)	Tier 2 immediately; advise driver to contact their bank
Fleet card issues	Direct to fleet administrator
Card suspended	Tier 2 review required to reinstate
Driver reports account was closed without their action	Tier 2 immediately — potential security incident

Agent Reminders

- Never update payment card details over the phone. Always direct to the app or web portal.

- Verify identity before making any account changes — name + email + last 4 of card on file is the minimum.
- If a driver says their account shows sessions they didn't initiate, treat this as a potential security incident — escalate to Tier 2 and advise them to contact their bank.
- RFID card technique is the most common fix. Don't skip it even if the driver sounds tech-savvy.
- Plug-and-charge is still not universal — not all vehicles or stations support it. Set correct expectations.

KB-008 — Stuck / Locked Cable

Category: Physical Cable Issues

Call Volume: ~4% of contacts

Urgency: HIGH — driver is physically unable to leave

Primary Remote Actions: Remote connector unlock, technician dispatch

Why This Matters

A stuck cable is one of the most stressful situations an EV driver can face — their vehicle is physically tethered to a charger they cannot leave. Resolution needs to be fast and clear. Importantly, agents must never advise forcing the cable out: the CCS and CHAdeMO connectors have mechanical locking latches, and forcing them can cause expensive damage to the vehicle's charging inlet (costs can exceed \$1,000–\$3,000 to repair).

Background: Why Cables Lock and Stick

The Locking Mechanism

DC fast charging connectors (CCS and CHAdeMO) engage a mechanical latch when a session is active. This latch serves two purposes: 1. **Safety** — prevents accidental disconnection during high-voltage DC charging 2. **Integrity** — ensures the connection remains stable throughout the session

The latch is controlled by: - The charger's internal connector lock mechanism (a small motor or solenoid) - The vehicle's inlet latch (some vehicles have a secondary lock, controlled by the vehicle's charging controller)

Both must release for the connector to come out. If either stays locked, the cable is stuck.

Why a Cable Gets Stuck

Cause	Frequency	Notes
Session still technically active (or ghost session)	~40%	Easiest fix — end session, then unlock
Charger software failed to send unlock signal at session end	~25%	Remote unlock resolves
Vehicle inlet latch did not release (vehicle-side)	~15%	Vehicle manual release or dealer
Mechanical failure in charger lock mechanism	~10%	Remote unlock may not work — dispatch
Power loss to charger during active session	~5%	Manual release on charger or dispatch
Connector deformed or wedged due to prior misuse	~5%	Do not force — dispatch

Diagnostic Script

Step 1 — Confirm the scenario

“Is the cable stuck in your car’s port, or stuck at the charger end?”

Almost always: stuck in the vehicle’s charging port (the connector won’t pull out of the car).

Step 2 — Confirm session status

“What does the charger screen show right now — does it say the session is still active, or has it ended?”

“What does your vehicle’s display show — is it still showing a charging indicator?”

If session is still active:

“The connector is locked because the session is still running — that’s actually the latch doing its job. Let me help you end the session properly first.”

End session via: 1. Ask driver to tap “Stop Session” on the charger screen 2. Or stop via the network app 3. If neither works, agent triggers a remote stop via the backend

After the session ends, the connector should release. Ask driver to try again.

If session shows as ended but cable is still stuck:

“I see the session has ended on our side too. The connector should have released — let me send a remote unlock command to the station. Give it about 10 seconds after I send it, then try the cable.”

Trigger remote connector unlock. Wait 10–15 seconds.

“Okay, try the cable now — use a firm but smooth pull, don’t yank it.”

Step 3 — If remote unlock succeeds

“Did it release? Perfect. Make sure the cable is fully out before driving — and let me know if there’s anything else I can help with.”

Step 4 — If remote unlock fails

“The remote unlock didn’t work — the locking mechanism may have a hardware fault. Let’s try a couple of alternatives before we dispatch a technician.”

Vehicle-side manual release (check vehicle-specific options below):

Consult the vehicle-specific notes in Section B below.

Power cycle the charger: > “I’m going to try a full power cycle on the station — sometimes that releases the lock. This takes about 90 seconds.”

Trigger hard reset. After reset, try remote unlock again.

If still stuck after hard reset and remote unlock:

“I’m going to dispatch a technician to help with this. I know that’s not the answer you wanted — I’m really sorry. Can I get your name and a callback number? They’ll aim to reach you within [X minutes per regional SLA].”

Section A — Driver Actions That Are SAFE

Agents may suggest these while waiting for a technician:

1. **Lock and unlock the vehicle** — some vehicle systems re-query the inlet status on unlock, which can trigger a release

2. **Power cycle the vehicle** — turn off completely (not just accessory mode), wait 30 seconds, power back on. This resets the vehicle’s charging controller and may release the inlet latch.
3. **Check in-vehicle settings** — some vehicles have a “Unlock charge port” or “Stop charging” option in the infotainment system or in the manufacturer’s companion app
4. **Open and close the vehicle’s charge port cover** (if motorized) — cycling it may reset the inlet latch signal
5. **Vehicle emergency cable release** — see Section B

Section A — Actions That Are NOT SAFE

Agents must never advise these:

- **Pulling forcefully on the cable** — can damage the vehicle’s CCS inlet, connector latch, or the charger cable. Inlet repair costs \$1,000–\$3,000+.
 - **Using a tool (pry bar, screwdriver) on the connector** — will damage both connector and inlet.
 - **Disconnecting the charger from mains power** (at the panel) — not the driver’s job; creates safety hazards; may not release the lock on all charger models.
 - **Leaving the vehicle and driving away** — physically impossible with the cable locked, and attempting it will cause severe damage.
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Section B — Vehicle-Specific Manual Release Options

These are general guidelines. Always verify with the vehicle’s owner’s manual for the specific model year.

Tesla (NACS connector)

“On Tesla vehicles, there’s a manual cable release. Hold the rear driver-side door handle for about 3 seconds — that triggers an emergency release on some models. Alternatively, go to the charging menu on the screen and tap ‘Open Charge Port.’”

Tesla NACS connectors lock differently from CCS — they use a pin-actuated latch. If the vehicle is powered off and unresponsive, there may be a physical pull-tab behind the rear license plate or in the trunk (model-specific).

Hyundai / Kia (CCS)

Emergency manual release is located in the trunk or cargo area: > “In the trunk of your vehicle, there should be an emergency cable release — it looks like a small loop or handle near the charge port side of the trunk lining. Pulling it manually releases the inlet latch. Check your owner’s manual for the exact location.”

Chevrolet Bolt / Bolt EUV (CCS)

Manual release is in the front trunk (frunk) area or trunk: > “There’s an emergency charge port release cable in your vehicle — usually in the front storage area or trunk. It’s a small pull-cord that releases the port latch manually. Your owner’s manual will have the exact location.”

Ford Mustang Mach-E / F-150 Lightning (CCS)

Manual release accessible from inside the cargo area or via the Ford Pass app: > “Try using the Ford Pass app on your phone — there’s an option to unlock the charge port. If that doesn’t work, there’s a physical emergency release in the cargo area.”

Rivian (CCS / NACS)

Rivian has an emergency charge port release accessible via the Rivian app or from inside the vehicle: > “Open the Rivian app and look for the charge port unlock option. If the app isn’t working, there’s a manual release — check your vehicle’s settings screen under Charging, or consult the Rivian owner’s guide.”

Volkswagen / Audi / Porsche (CCS2 in Europe; CCS1 in US)

Some models have a manual release accessible from inside the vehicle: > “Check your vehicle’s infotainment system under Vehicle Settings > Charging — there may be an option to manually unlock the charge port. Your owner’s manual will also have a physical emergency release location.”

Nissan (CHAdMO on older models, CCS on newer)

“For CHAdMO vehicles like the LEAF, the charging port has a button on the charge port cover that can release the cable once the session is stopped. Try pressing the charge port release button on the vehicle while the session is confirmed stopped.”

Section C — Charger-Side Physical Release (Technician Action Only)

Most modern DC fast chargers have a manual override for the connector lock accessible to technicians — typically a small lever or button inside the charger cabinet behind a service panel. This is not accessible to drivers and requires a trained technician.

Do not instruct drivers to attempt to open the charger cabinet — it contains high-voltage equipment and is not safe for untrained personnel.

Section D — If the Driver Must Leave Urgently

If the driver has an emergency and cannot wait for a technician:

1. Confirm whether the vehicle has a tow mode or transport mode that can be activated while tethered (most do not, but check)
2. Contact local roadside assistance — some EV roadside services carry connector release tools
3. Contact the vehicle manufacturer’s roadside assistance line — they may have model-specific guidance
4. In absolute last resort, and only if the driver or a roadside professional does it: some connectors can be released by inserting a thin tool to manually depress the latch — this risks damage and should only be done by someone who knows the specific connector mechanism

“I strongly recommend waiting for our technician if you can — forcing the cable risks expensive damage to your car’s charging port. But if you need emergency help sooner, your vehicle manufacturer’s roadside line may be able to assist.”

Escalation Criteria

Condition	Action
Remote unlock fails after hard reset	Dispatch technician — priority
Driver has emergency and cannot wait	Contact regional tech dispatch for fastest ETA; also advise vehicle manufacturer roadside
Connector or inlet appears physically damaged	Dispatch + advise driver not to drive until inspected
Charger lost power during session (cable locked, charger off)	Dispatch — manual release required on charger side

Billing Note

If a driver was billed for time their vehicle was stuck at the charger (idle fees in particular), flag the session for billing review:

“I’ll make sure our billing team looks at this session — you won’t be charged idle fees for time the cable was stuck. Can I confirm your email for the review notification?”

Agent Reminders

- Stuck cable is an urgent situation — treat it with priority and empathy. The driver is stranded.
- Always try ending the session first before sending the unlock command.
- Never advise forcing the connector. The repair cost for a damaged CCS inlet can exceed the value of a month of charging.
- If dispatching a tech, give a realistic time estimate based on regional SLA — don’t say “shortly” without knowing.
- Log vehicle make/model for the technician — release procedures vary by vehicle.

KB-009 — Safety Escalation Protocols

Category: Safety — ALWAYS ACTIVE

Call Volume: Low — but highest priority of any issue

Authority: Any agent can and must escalate safety concerns immediately

Primary Actions: Immediate disconnect instruction, emergency services referral, safety dispatch

Critical Rule

Safety escalations override all other call flows. If a driver mentions any of the safety triggers listed below — mid-troubleshooting, mid-billing dispute, at any point — stop the current flow and go to the appropriate safety script immediately.

Safety Trigger Recognition

Escalate immediately if the driver mentions any of the following:

Trigger phrase / description	Category	Response
“Sparks,” “sparking,” “arc,” “flash of light”	Electrical arc	Section A
“Burning smell,” “smoke,” “it smells like something’s burning”	Thermal event / fire risk	Section A
“Fire,” “flames,” “it’s on fire”	Active fire	Section A — call emergency services
“I got a shock,” “it shocked me,” “electric shock,” “tingling”	Electrical contact	Section B
“The connector is melted,” “the cable is melted,” “it’s discolored and hot”	Thermal damage	Section C
“My car is making a strange noise from the battery,” “my car smells like sulfur”	Battery thermal event (vehicle)	Section D
“The charger fell over,” “it looks structurally damaged,” “the cabinet is open”	Physical safety hazard	Section E
“There’s water in the charging port,” “flooding,” “standing water near the charger”	Water / electrical hazard	Section F
“My car battery swelled,” “puffed up,” “the battery looks wrong”	Battery deformation	Section D

Section A — Electrical Arc, Smoke, or Fire

This is the highest-priority scenario.

Script (read verbatim if possible)

“I need you to stop what you’re doing right now. Please do not touch the connector or the charger. If the cable is plugged in and safe to reach without getting near the sparking area, unplug it — but only if it’s safe. Otherwise, step away from the charger and your vehicle immediately. Get at least 30 feet away.”

If there are visible flames or active smoke:

“Please call 911 immediately — do not wait. Tell them there is an electrical fire at an EV charging station. Then step away from the vehicle. Do not attempt to put out an EV battery fire with a conventional extinguisher — those fires require special suppression. Get to safety and let the fire department handle it.”

Agent actions (immediately after driver is safe)

1. **Flag the station as SAFETY EMERGENCY** in the backend — immediately taken offline, no remote resets, no other drivers can start sessions
2. **Dispatch a safety-priority technician** — not a standard tech ticket. This is a priority safety inspection.
3. **Log the incident** with: station ID, date/time, driver name and contact, description of what was observed, whether driver was injured
4. **Notify the on-call supervisor** — all safety incidents involving arc, fire, or smoke require supervisor notification within 15 minutes
5. **Do not reset or unlock anything** on this station until a technician has physically inspected it

If driver reports they’ve already left the area

“That was the right call. Is everyone okay? Are you safe right now?”

Confirm safety, then proceed with logging. The station is already flagged. Advise the driver that a technician will inspect the station and they may be contacted for more details if an incident report is needed.

Section B — Electric Shock

Electric shock from an EV charger can range from a mild static-like sensation to a serious injury. Treat all shock reports seriously.

Immediate script

“I’m very sorry to hear that. First — are you feeling okay right now? Are you in pain, feeling dizzy, or having any unusual sensations?”

If driver reports significant pain, difficulty breathing, numbness, muscle spasms, or any signs of serious injury:

“Please call 911 immediately. Electric shock can have internal effects that aren’t immediately obvious. Please get medical attention right now — I can stay on the line with you. Do not drive if you feel unwell.”

If driver reports only a mild, brief sensation (similar to static):

“I understand — are you feeling okay now? Any tingling or numbness?”

If they feel fine: > “I’m glad you’re okay. Please don’t touch the connector or the charger again — I’m flagging this station for a safety inspection right now.”

Agent actions

1. **Flag station as SAFETY HOLD** in the backend immediately
2. **Log incident:** station ID, time, driver name and contact, description of the shock (mild/severe, where on body, duration), any reported injury
3. **Dispatch technician** for electrical safety inspection — ground fault testing required
4. **Notify supervisor** — all shock incidents require supervisor notification
5. **Document driver’s stated condition** — if they decline emergency services but later develop symptoms, this log is important

Common causes of charger-related shock sensations

- **Ground fault** at the charger (the most serious — triggers E005/GFCI error; charger should self-disable)
 - **Static electricity** — not from the charger itself, but from the vehicle or environment; harmless but startling
 - **Damaged cable insulation** — a serious safety fault; station must be taken offline immediately
 - **Faulty vehicle inlet** — if the vehicle’s inlet has insulation damage, the shock originates at the vehicle; advise vehicle inspection
-

Section C — Melted or Thermally Damaged Connector / Cable

If a driver describes a connector that appears melted, discolored, deformed, or extremely hot to the touch:

Script

“Please do not touch the connector again — it may still be at unsafe temperatures, and the damage could indicate an ongoing electrical issue. Step away from the charger.”

If the connector is still in the vehicle:

“Do not try to remove it. A technician will need to do that safely. Is the session showing as ended on the screen?”

If session is ended and cable is not in the vehicle: advise the driver not to pick up the connector.

Agent actions

1. **Flag station as SAFETY HOLD** — immediately taken offline
2. **Dispatch priority technician** with a note: “Thermally damaged connector — do not restart until inspection”
3. **Advise driver to inspect their vehicle’s charging inlet** for signs of heat damage — if they see discoloration, they should not charge the vehicle again until the inlet is inspected by their dealer
4. **Log:** station ID, connector description, driver contact info, vehicle make/model
5. **Notify supervisor**

Why connector overheating happens

- **Loose connection** — a slightly unseated connector creates resistance, which generates heat; repeated over many sessions, this can cause progressive thermal damage
 - **Pin wear / corrosion** — worn or corroded connector pins create resistance at contact points
 - **Overcurrent fault** — charger delivers more current than the cable rating; hardware fault
 - **Damaged cable insulation** — exposes conductors; risk of arc and fire
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Section D — Vehicle Battery Thermal Event

Lithium-ion batteries can undergo “thermal runaway” — a self-sustaining exothermic reaction that generates intense heat and fire. This is rare but extremely dangerous. EV battery fires burn very hot, can reignite after appearing extinguished, and require massive water application to cool (not conventional extinguishers).

Trigger signs (driver describes)

- Swelling or deformation of the vehicle body near the battery
- Sulfur or “rotten egg” smell coming from the vehicle
- Smoke coming from under the vehicle
- Battery warning light plus unusual heat
- Crackling, hissing, or popping sounds from under the vehicle floor

Script

“Please get away from the vehicle immediately — at least 100 feet if possible. Do not get back in the car. Call 911 right now and tell them you have a potential EV battery fire. Tell them the vehicle make, model, and color.”

“Are you able to get away from the vehicle safely?”

Confirm the driver is moving to safety. Stay on the line if possible.

“The fire department will need a lot of water to cool a lithium battery fire — they’re trained for this. Please don’t attempt to put it out yourself. I’m flagging this as an emergency on our end.”

Agent actions

1. **This is a 911 emergency** — confirm driver has called or is calling
 2. **Flag station as emergency shutdown** — do not allow any use until fire department and technician have cleared it
 3. **Immediately notify supervisor** — this is a critical incident requiring executive notification per most network incident response plans
 4. **Do not remotely reset, unlock, or take any remote action** on the station until cleared by emergency responders
 5. **Document:** time, station ID, vehicle details, driver contact, nature of the event
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Section E — Physical Damage to Charger Structure

If a driver reports the charger has fallen over, the cabinet is open or damaged, or there is physical damage exposing internal components:

Script

“Please step away from the charger — don’t touch it. Exposed internal components can be at high voltage even when the station appears off.”

“Is the cable still plugged into a vehicle?”

If cable is in vehicle: > “Don’t try to remove the cable. I’m dispatching a technician right now.”

If cable is not in vehicle: > “Leave the cable where it is and stay away from the unit. I’m flagging it now.”

Agent actions

1. **Flag station as SAFETY HOLD**
 2. **Dispatch technician** — note: physical damage, possible exposed internals
 3. **Log:** description of damage, whether cable is still in a vehicle, station ID, driver contact
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Section F — Water / Flooding Near Charger

EV charging stations are rated for outdoor use and weather exposure (typically IP54 or higher), but standing water, flooding, or direct water entry into the unit is a safety concern.

Script

“If there’s standing water near or under the charger, please don’t approach it. Water and electrical equipment can be a dangerous combination even with safety ratings.”

If the charger is clearly submerged or flooded: > “Do not approach the charger at all. I’m flagging this station for emergency inspection.”

If the water is minimal (puddle from rain, not flooding): > “Modern chargers are designed to handle rain and wet conditions, but I’m going to flag this one for inspection to be safe. Is the charger showing any error messages?”

Post-Incident Documentation

All safety incidents require a complete log before the call ends. Collect:

Field	Required
Station ID	Always
Date and time of incident	Always
Driver name	Always
Driver contact (phone, email)	Always
Vehicle make, model, year	Always
Description of what was observed	Always
Whether emergency services were contacted	Always
Driver’s reported physical condition	Always
Whether driver accepted or declined medical attention	Always
Remote actions taken (if any)	Always
Technician dispatch confirmed	Always
Supervisor notified	Always

Supervisor Notification Thresholds

Incident type	Notification timeline
Electric shock — any severity	Within 15 minutes
Arc, sparking, fire	Within 15 minutes

Incident type	Notification timeline
Vehicle battery thermal event	Immediately
Thermally damaged connector	Within 30 minutes
Physical damage with exposure risk	Within 30 minutes
Driver reported injury (any)	Within 15 minutes

Agent Reminders

- **Safety overrides everything.** Do not finish a billing script, a session reset, or any other flow if a safety trigger is present.
- **Confirm the driver is safe before logging.** The log can wait. The driver cannot.
- **Never reset a station that has a safety flag.** Even if a driver requests it.
- **Do not minimize a shock report.** What feels like “just a tingle” can mask internal injury. Always recommend medical attention for any reported shock.
- **EV battery fires are not like other fires.** A vehicle fire must be handled by trained fire department — never advise a driver to use a fire extinguisher on a battery fire.
- **You have the authority to dispatch and flag.** You do not need supervisor approval to take a station offline for safety. Act first, notify after.